

Executive Summary

SIOS Sales Swarm is an innovative multi-agent system (MAS) designed to replace traditional door-to-door (D2D) sales with high-conversion digital engagement. Targeted at home service industries—including HVAC, Solar, Roofing, and Pest Control—the platform deploys a collaborative "swarm" of AI agents. Each agent is endowed with a unique digital persona, specialized skill set, and synchronized lifelike avatar to build immediate trust with prospects.

UI/UX & Digital Persona Innovation

To achieve "market-friendly" interactions, the platform prioritizes human-centric design and psychological safety.

- **Synthetic Media Integration:** Leveraged cutting-edge APIs to create synchronized, real-time interactive avatars that speak with distinct, emotionally intelligent voices.
- **Persona Builder UI:** Developed an intuitive configuration interface for SMBs to select agent personalities (e.g., "Enthusiastic," "Analytical," or "Empathetic") and assign them unique backstories.
- **Interactive Web Widgets:** Designed embeddable WebRTC-based widgets where prospects engage in real-time voice or text conversations with AI personas.
- **Ethical Disclosure Design:** Implemented transparent "AI-Generated" signaling to maintain prospect trust and comply with emerging synthetic media regulations.

Performance & Optimization Standards

The website and platform components adhere to the highest modern web standards for conversion and speed.

- **Mobile-First Design:** All prospect-facing landing pages and video interfaces are fully responsive and touch-optimized.
- **Low-Latency Pipeline:** Engineered a real-time conversational AI pipeline (STT → LLM → TTS → Avatar) utilizing WebSockets for near-instantaneous response times.
- **Secure Payment Ecosystem:** Integrated a robust Stripe processing layer featuring encrypted API management, HTTPS enforcement, and a frictionless checkout flow.

- **Conversion Optimization:** Utilized A/B testing on agent personas and outreach cadences to identify the most effective engagement strategies for specific industries.

Tooling & Tech Stack

- **AI Orchestration:** LangChain, CrewAI.
- **Synthetic Media:** ElevenLabs (Voice), HeyGen (Interactive Avatar).
- **Real-time Comms:** WebRTC, LiveKit, WebSockets.
- **Infrastructure:** Cloud-native microservices, Vector Databases (Pinecone/Weaviate), Stripe API.